

# Georgios Palkanoglou

## Contact

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## Current position

*Postdoctoral Researcher*, TRIUMF Canada's Particle Accelerator Centre, Vancouver, Canada

## Education

- 2019–2023 Ph.D. in Physics, University of Guelph  
Dissertation: *Pairing in Nuclear and Cold Atomic Systems*
- 2018–2019 M.Sc. in Physics, University of Guelph  
Thesis: *Approaching the Thermodynamic Limit in BCS Theory*
- 2013–2017 B.Sc. (HONOURS) in Physics, National and Kapodistrian University of Athens

## Awards & Honours

- 2025–2026 Catalonia Quantum Academy Visiting Scholar Award
- 2025 Canadian Institute for Nuclear Physics support for workshop organization
- 2020 Constantine and Patricia Mavroyannis Scholarship
- 2019–2020 Dean's Graduate scholarship (University of Guelph)
- 2019 International Doctoral Training Scholarship - Ph.D.'s (University of Guelph)

## Conference organizing

- 2027 (*Upcoming*) Lead organizer of the ECT\* workshop "*Bridging Effective Field Theory Across Subatomic Scales*"
- 2025 Lead organizer of the ESNT workshop (**with CINP support**) "*Ab initio many-body calculations: where has the pairing gone?*"
- 2025 Lead organizer of the TRIUMF workshop "*SMEFT meets ChEFT*"
- 2025 Co-organizer of the TRIUMF workshop "*PAINT26: Progress in Ab Initio Nuclear Theory*"

## Spokesperson of Collaboration

- 2025 S1936 at TRIUMF: *Beam Development of Light Lanthanides for Nuclear Structure Investigations Approaching  $N = Z$*

## Teaching experience

- 2026 Instructor for "*Ab initio nuclear theory: a crash course*", University of Barcelona, Barcelona, Spain
- 2023 Graduate Teaching Assistant in Physics of Music, Department of Physics, University of Guelph
- 2021 Graduate Teaching Assistant in Quantum Mechanics, Department of Physics, University of Guelph
- 2020 Graduate Teaching Assistant in (graduate) Quantum Field Theory, Department of Physics, University of Guelph
- 2019–2023 Private Tutor in Undergraduate Physics
- 2018–2023 Graduate Teaching Assistant in Applied Physics, Department of Physics, University of Guelph
- 2013–2017 Private Tutor in Physics, Maths, and Geometry

## Student co-Supervision

- 2026– Isiah Bennet, M.Sc. student, University of Guelph  
2024–2025 Christos Tsekouras, Undergraduate thesis, National and Kapodistrian University of Athens  
2021–2022 Michael Stuck, Undergraduate Co-op student, University of Guelph  
2022 Kareena Bhalla, Undergraduate Co-op student, University of Guelph

## Publications

### JOURNAL ARTICLES

- 2026 *Exploring the shell structure of trapped superfluids*  
G. Palkanoglou, R. Curry, and S. Gandolfi  
[arXiv:2607.14214](#) [nucl-th]
- 2026 *Interplay between Nuclear Shell Structure and Pairing around Doubly Magic  $^{132}\text{Sn}$*   
R. Simpson, G. Palkanoglou, E.C. Brisley, J.D. Cardona, A. Czhaly, S. Kakkar, M. Simonov, E. Taylor, C. Walls, P. Weligampola, C. Chambers, F. Maldonado Millan, A. Mollaebrahimi, D. Ray, A. Weaver, J. Yu, I. Dillmann, A. Gezerlis, G. Gwinner, A.O. Macchiavelli, S. Malbrunot-Ettenauer, M.P. Reiter, and A.A. Kwiatkowski  
[arXiv:2607.05647](#) [nucl-ex]
- 2026 *Evidence for Mutlimodal Superfluidity of Neutrons*  
Y. Ma, G. Palkanoglou, J. Carlson, S. Gandolfi, A. Gezerlis, G. Given, A. Hicks, D. Lee, K. E. Schmidt, and J. Yu  
[arXiv:2602.17611](#) [nucl-th]
- 2025 *A novel way of recasting the Bardeen-Cooper-Schrieffer gap equations*  
G. Palkanoglou and A. Gezerlis  
(in press at *Phys. Rev. C*) | [arXiv:2510.07384](#) [nucl-th]
- 2025 *Symmetry properties of pairing correlations in heavy deformed nuclei*  
G. Palkanoglou and A. Gezerlis  
[arXiv:2505.08879](#) [nucl-th]
- 2024 *Spin-Triplet Pairing in Heavy Nuclei is Stable Against Deformation*  
G. Palkanoglou, M. Stuck, and A. Gezerlis  
*Phys. Rev. Lett.* **134**, 032501 (2025) | [arXiv:2402.13313](#) [nucl-th]
- 2022 *The  $^1S_0$  pairing gap in neutron matter*  
S. Gandolfi, G. Palkanoglou, J. Carlson, A. Gezerlis, and K. E. Schmidt  
*Condens. Matter* **2022**, 7(1), 19 | [arXiv:2201.01308](#) [nucl-th]  
\*Marked as a Feature Paper
- 2021 *Superfluid neutron matter with a twist*  
G. Palkanoglou and A. Gezerlis  
*Universe* **2021**, 7(2), 24 | [arXiv:2012.04663](#) [nucl-th]  
\* Marked as a Feature Paper
- 2020 *From odd-even staggering to the pairing gap in neutron matter*  
G. Palkanoglou, F. K. Diakonou, and A. Gezerlis  
*Phys. Rev. C* **102**, 064324 (2020)  
[arXiv:2005.05985](#)

### PROCEEDINGS

- 2019 *Effective mass and pairing gap in neutron matter*  
G. Palkanoglou, N. Ismail, M. Buraczynski, and A. Gezerlis  
*J. Phys.: Conf. Ser.* **1643** 012132 (2020).

## Invited Talks & Seminars

- 2026 *Beyond the conventional nuclear superfluidity: triplet and mixed-spin superfluids across nucleonic systems, pairing at the drip-line, and more*  
University of Barcelona, Barcelona, Spain

- 2026 *Revisiting superfluidity and shell-effects at the crossroads: experiments, mean-field, and ab initio descriptions*  
UPC Universitat Politècnica de Catalunya, Barcelona, Spain
- 2026 *Nuclear superfluidity and clustering, many-nucleon forces, and ML methods for ab initio nuclear theory*  
Job Interview (virtual), ECT\*, Trento, Italy
- 2026 *Unconventional nuclear superfluidity: from the proton to the neutron drip-line, and beyond*  
Job Interview (virtual), University of Jyväskylä, Jyväskylä, Italy
- 2026 *(Co-organized) Beyond the leading three-nucleon force in light nuclei*  
PAINT<sub>26</sub>, Vancouver, BC, Canada
- 2025 *(Organized) Pinning the subleading three-nucleon contact interaction to light nuclei*  
SMEFT meets ChEFT workshop, TRIUMF, Vancouver, BC, Canada
- 2025 *(Organized) Ab initio calculations of singlet and triplet pairing gaps in nucleonic matter*  
Introductory lecture at Ab initio many-body calculations: “where has the pairing gone?”, ESNT CEA, Saclay, Paris, France
- 2025 *How to describe isoscalar pairing in heavy deformed nuclei and measure its effects*  
ACOT meeting, TRIUMF, Vancouver, BC, Canada
- 2025 *Putting the subleading three nucleon force to use*  
PAINT<sub>25</sub>, TRIUMF, Vancouver, BC, Canada
- 2025 *S1936 Progress Report: Beam Development of Light Lanthanides for Nuclear Structure Investigations Approaching  $N = Z$*   
NP-EEC, TRIUMF, Vancouver, BC, Canada
- 2024 *Exotic pairing phases in nuclei and neutron matter*  
T-2 Seminar, LANL, Los Alamos, NM, USA
- 2024 *Exotic nuclear superfluidity in heavy nuclei*  
3rd APCTP-TRIUMF joint workshop, PKNU, Busan, South Korea
- 2024 *Nuclear superfluidity in nuclei and neutron matter*  
National and Kapodistrian University of Athens, Athens, Greece
- 2024 *(Pathways to) Exotic pairing in heavy nuclei*  
PAINT<sub>24</sub> - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, BC, Canada
- 2023 *Novel pairing and testing EFT interactions in neutron matter and nuclei*  
Michigan State University
- 2023 *Mixed-spin pairing in nuclear and cold atomic systems*  
(Virtual) Warsaw University of Technology  
(Job Interview)
- 2023 *Mixed-spin pairing in nuclear and cold atomic systems*  
(Virtual) TRIUMF Theory Group  
(Job Interview)
- 2023 *Representing sound: Acoustics & Phonetics*  
One lecture in Phonetics  
School of Languages and Literature, University of Guelph, Guelph, ON, Canada

## Contributed talks

- 2026 *Superfluid effects in deformed heavy nuclei close to the drip-lines*  
Nuclear Theory in Spain I, Universidad Complutense de Madrid, Madrid, Spain
- 2024 *Exotic nuclear superfluidity in heavy nuclei*  
2024 CeNAM Frontiers, University of Notre Dame, South Bend, IN, USA
- 2024 *Exotic nuclear superfluidity in heavy nuclei*  
Theory Canada 16, Western University, London, ON, Canada
- 2024 *Exotic nuclear superfluidity in heavy nuclei*  
CAP Congress 2024, Western University, London, ON, Canada
- 2023 *Singlet and triplet pairing in nuclear and cold atomic systems*  
2023 Canadian Association of Physicists Congress

|      |                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | University of New Brunswick, Fredericton, NB, Canada                                                                                                                                                              |
| 2023 | <i>Singlet and triplet pairing in nuclear and cold atomic systems</i><br>60 <sup>th</sup> Winter Nuclear and Particle Physics Conference<br>Banff, AB, Canada                                                     |
| 2023 | <i>Mixed-spin pairing in nuclear and cold atomic systems</i><br>Toronto Ultracold Atom Network (TUCAN) meeting<br>University of Toronto, Toronto, ON, Canada                                                      |
| 2022 | <i>The S-wave pairing gap in neutron matter</i><br>2022 Canadian Association of Physicists Congress<br>*Marked as the 3 <sup>rd</sup> best presentation in the Division of Theoretical Physics (DTP)              |
| 2021 | <i>Superfluid neutrons: from particles to matter</i><br>2021 Canadian Association of Physicists (Virtual) Congress<br>*Tied as the 2 <sup>nd</sup> best presentation in the Division of Theoretical Physics (DTP) |
| 2021 | <i>Superfluid neutron matter with a twist</i><br>58 <sup>th</sup> Winter Nuclear and Particle Physics (Virtual) Conference<br>* Marked as the 2 <sup>nd</sup> best presentation in Theoretical Physics            |
| 2021 | <i>Superfluid neutron matter with a twist</i><br>(Virtual) Graduate Seminar Series<br>University of Guelph, Guelph, ON, Canada                                                                                    |
| 2020 | <i>From odd-even staggering to the pairing gap in neutron matter</i><br>2020 Canadian Association of Physicists (Virtual) Congress                                                                                |
| 2020 | <i>From odd-even staggering to the pairing gap in neutron matter</i><br>57 <sup>th</sup> Winter Nuclear and Particle Physics Conference<br>Banff, AB, Canada                                                      |
| 2018 | <i>A Superconducting Star</i><br>Graduate Seminar Series<br>University of Guelph, Guelph, ON, Canada                                                                                                              |

## Conferences & Summer schools attended

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|------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | ML2026 - II Machine Learning Methods for Complex and Quantum Systems, University of Barcelona, Barcelona, Spain                                 |
| 2026 | NT.ES I - Nuclear Theory in Spain, Universidad Complutense de Madrid, Madrid, Spain                                                             |
| 2026 | CSQCD - Compact Stars in the QCD phase diagram, University of Barcelona, Barcelona, Spain                                                       |
| 2026 | PAINT26 - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, British Columbia, Canada                                         |
| 2025 | SMEFT meets ChEFT, TRIUMF, Vancouver, British Columbia, Canada                                                                                  |
| 2025 | Ab initio many-body calculations: where has the pairing gone, ESNT, Saclay, France                                                              |
| 2025 | PAINT25 - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, British Columbia, Canada                                         |
| 2024 | 3rd APCTP-TRIUMF joint workshop: From Nuclei to Neutron Stars, PKNU, Busan, South Korea                                                         |
| 2024 | 2024 CeNAM Frontiers, University of Notre Dame, South Bend, Indiana, USA                                                                        |
| 2024 | 2024 C.A.P Congress, Western University, London, Ontario, Canada                                                                                |
| 2024 | Theory Canada 16, Western University, London, Ontario, Canada                                                                                   |
| 2024 | PAINT24 - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, British Columbia, Canada                                         |
| 2023 | 2023 C.A.P Congress, University of New Brunswick, Fredericton, New Brunswick, Canada                                                            |
| 2023 | 60 <sup>th</sup> Winter Nuclear and Particle Physics (Virtual) Conference, Banff, AB, Canada                                                    |
| 2022 | 2022 C.A.P Congress, McMaster University, Hamilton, Ontario, Canada                                                                             |
| 2021 | 2021 C.A.P (Virtual) Congress                                                                                                                   |
| 2021 | 58 <sup>th</sup> Winter Nuclear and Particle Physics (Virtual) Conference                                                                       |
| 2020 | 2020 C.A.P (Virtual) Congress                                                                                                                   |
| 2020 | 57 <sup>th</sup> Winter Nuclear and Particle Physics Conference, Banff, AB, Canada                                                              |
| 2017 | The 2017 Onassis Foundation Lectures in Physics: “Quantum physics frontiers explored with cold atoms, molecules and photons”, Heraklion, Greece |
| 2017 | The 2017 Petnica Summer Institute summer school on Cosmology, Petnica, Serbia                                                                   |
| 2016 | The 3 <sup>rd</sup> Xmas Theoretical Physics Workshop, Athens, Greece                                                                           |

## Computing skills

### LANGUAGES

Fortran, Python, Bash, MATLAB, Mathematica, C++, HTML

### SOFTWARE

L<sup>A</sup>T<sub>E</sub>X, PLUTO, Logger Pro

## Outreach & Service

- 2025 Volunteer mentor in the Early Career Researcher mentorship program, TRIUMF
- 2022 Research Information Session, University of Guelph
- 2021 Volunteer judge for the 57th Annual Canadian Undergraduate Physics Conference (CUPC)
- 2019 Volunteer for STEM Week 2019 organized by the University of Guelph, Guelph Civic Museum
- 2018 Volunteer for STEM Week 2018 organized by the University of Guelph, Guelph Civic Museum

## Languages

Greek (native), English (fluent), Farsi (beginner), Spanish (beginner), German (beginner)